Does distress risk language in AMD's Item 1A (Risk Factors) increase before periods of stock price decline

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1. Introduction & Research Question

In recent years, AMD has experienced significant cyclical fluctuations and strategic transformation within the semiconductor industry. During the period from 2012 to 2016, the company faced persistent losses, cash flow pressures, and rising debt burdens, before achieving a phased recovery through product innovation and market expansion. However, the semiconductor industry is highly dependent on macroeconomic conditions, the pace of technological advancement, and access to capital markets, making firms' operating performance inherently volatile. In this context, whether the risk factors disclosed in AMD's 10-K filings are able to reflect potential operational and financial pressures in advance becomes a question worthy of systematic investigation.

This study focuses on AMD as the research sample and examines the Item 1A (Risk Factors) section of its annual 10-K reports. We construct a measure of financial distress-related risk language and test whether such risk disclosures intensify prior to significant stock price declines. By doing so, the study seeks to understand whether firms release forward-looking signals through risk disclosures before formal financial deterioration becomes evident, thereby providing empirical evidence for investor risk identification, expectation formation, and the assessment of disclosure quality.

2. Data

2.1 Text Data: AMD 10-K Item 1A (2011–2024)

We collect AMD annual 10-K filings from 2011 to 2024 (14 annual observations). For each filing, we extract Item 1A (Risk Factors) and apply the retrieval pipeline described in Section 3.

For each filing year, we construct three comparable distress indices (Hard, Soft, and Operational) in a globally fitted TF-IDF space. These indices summarize the prevalence and strength of distress-related language in Item 1A disclosure.

2.2 Stock Price and Market Benchmark Data

We use AMD total return index data and the S&P 500 total return index (SPX) as the market benchmark. Monthly returns are computed for both series.

Abnormal return is defined as the market-adjusted return:

$$Abnormal\ Return_t = r_t^{AMD} - r_t^{SPX}$$

To evaluate whether distress language predicts subsequent performance, we construct **forward cumulative abnormal returns** over 6-month and 12-month horizons by **summing monthly**

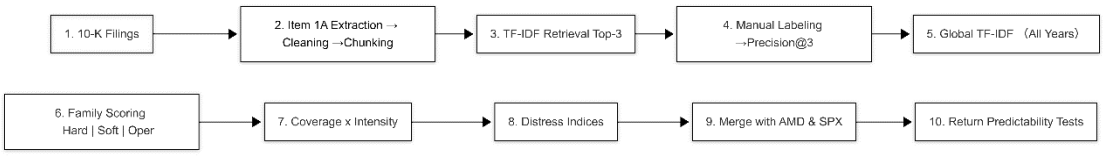
abnormal returns:

$$FwdAbRet_{t,h} = \sum_{i=1}^{h} AbRe\, t_{t+i}, h \in \{6,12\}$$

This rolling-sum formulation provides a simple and transparent approximation of cumulative abnormal performance and matches our empirical implementation.

3. Methodology

3.1 Workflow diagram



The diagram provides an overview of the analytical workflow. For clarity, only the main stages are shown; detailed implementation steps are documented in the accompanying code.

3.2 Item 1A extraction

Item 1A (Risk Factors) is extracted using rule-based pattern matching. Specifically, we locate the second occurrence of "ITEM 1A. RISK FACTORS" in the filing text (to exclude the table of contents) and truncate the document at the beginning of Item 1B/Item 2. This ensures that only the substantive risk disclosure section is retained for analysis.

3.3 Chunking strategy

Chunking is performed within each extracted Item 1A text by splitting on whitespace and forming overlapping word windows. We use a window length of 110 words and an overlap of 30 words. Each chunk is tagged with its filing year and a unique identifier to ensure traceability.

3.4 Retrieval model (TF-IDF + cosine ranking)

Item 1A chunks were initially indexed separately for each filing year using a year-specific **TfidfVectorizer**. To ensure cross-year comparability, we subsequently refit a single global TF-IDF model on the full set of chunks (2011–2024). All distress indices and empirical results reported in Section 5.1 are based on this globally fitted representation.

Given a distress query, similarity scores are computed using cosine similarity within the shared TF-IDF space, and chunks are ranked accordingly. The top-k highest-scoring chunks are returned as evidence candidates.

3.5 Query families (Hard / Soft / Operational)

Distress language is defined using three predefined query families: Hard, Soft, and Operational. Each family consists of a fixed set of keyword queries. For each family, similarity scores are computed between all Item 1A chunks and the corresponding query set within the global TF-IDF space. The complete query lists are provided in **Appendix A**.

In earlier exploratory analysis, we also examined a simpler index based on a single distress query set fitted separately by year. However, because such per-year TF-IDF representations are not comparable across time, this exploratory metric is not used in the final empirical analysis.

3.6 Index construction (Coverage × Intensity)

For each chunk, a family score is defined as the maximum similarity across the queries within that family.

- **Coverage:** the proportion of chunks whose scores exceed a global threshold.
- **Intensity:** the average score among those high-scoring chunks.
- **Index:** Coverage × Intensity.

This formulation incorporates both the prevalence and the magnitude of distress-related language within each filing, while reducing sensitivity to isolated high-scoring chunks.

3.7 Market Tests

The annual distress indices are merged with forward abnormal return measures defined in Section 2 ($\text{Abnormal Return}_t = r_t^{AMD} - r_t^{SPX}$). We assess the association between each index and subsequent performance using Spearman rank correlations, median-split comparisons, and simple OLS regressions. All tests are conducted on the annual time series (2011–2024).

3.8 Empirical Visualization

To provide graphical support for the statistical results, we present three complementary visualizations. First, a scatter plot of the distress index against forward 12-month abnormal returns with a fitted regression line. Second, a time-series comparison of the normalized distress index and subsequent abnormal returns. Third, a median-split bar chart comparing average forward returns between high- and low-distress years.

These figures illustrate the direction and relative magnitude of the estimated relationships.

4. Evaluation

4.1 Labeling design

To evaluate the quality of the retrieval layer, we manually label a stratified sample of retrieved chunks. Specifically, for each query, eight chunks are randomly sampled from the retrieval output, resulting in a total of 96 labeled observations.

Each sampled chunk is assigned a binary relevance label (1 = substantively relevant to the corresponding query; 0 = not substantively relevant). Evaluation focuses on whether highly ranked chunks genuinely reflect the intended distress concept. This sampling strategy ensures that all query families are represented in the evaluation set, rather than concentrating only on the highest-scoring queries. The complete list of distress queries is provided in **Appendix B**.

4.2 Precision@k

Retrieval performance is evaluated using Precision@k with k=3.
Precision@3 is defined as:

$$Precision@3 = \frac{Number\ of\ relevant\ chunks\ among\ ranks\ 1-3}{Total\ number\ of\ chunks\ ranked\ 1-3}$$

Across the 96 labeled observations:

- Relevant chunks: 22
- Precision@3 = 0.229

This indicates that approximately 23% of the top-ranked retrieval results are substantively relevant, suggesting that while relevant evidence appears in the highest ranks, retrieval noise remains non-trivial.

4.3 Query-level evaluation (what works / what fails)

Precision varies substantially across queries. Queries associated with concrete financial events, such as default events and goodwill impairment. In contrast, several covenant-related and more general distress queries yield no relevant chunks within the labeled sample.

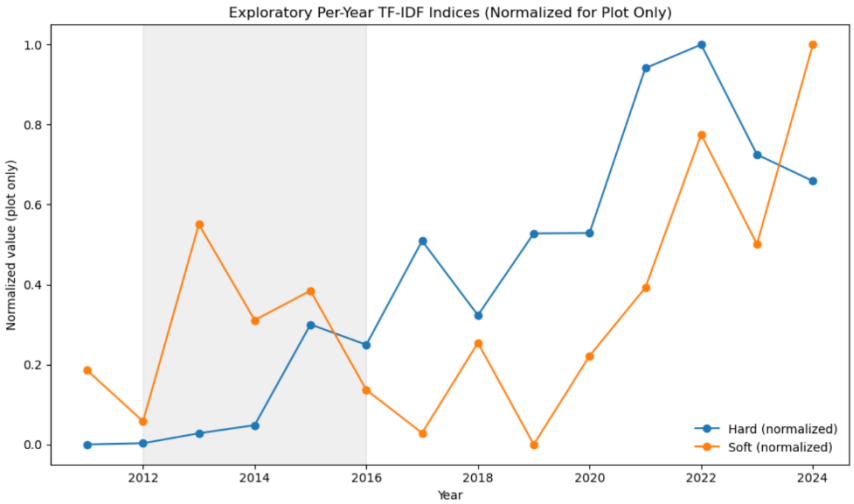
This pattern suggests that retrieval performance depends on the specificity and phrasing of the query. Precision varies across queries, with some terms achieving substantially higher precision than others. Detailed results are reported in **Appendix C**.

5. Results

5.1 Distress index visualization

First, based on the index constructed using a year-by-year TF-IDF fitting approach, the results are shown in the figure below. During the period from 2012 to 2016, both indices remained at relatively low to moderate levels. Soft Distress was comparatively more pronounced, but did not exhibit a clear upward or downward trend. Although Hard Distress displayed a more visible increasing pattern, its overall magnitude remained limited, making it difficult to conclude whether risk disclosures intensified prior to significant stock price declines.

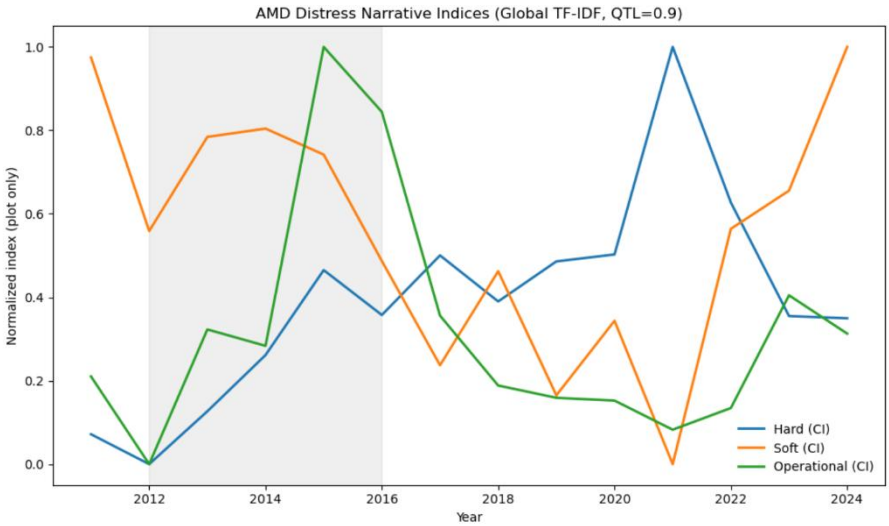
However, since these results are derived from a year-by-year fitting procedure, the vector space and weighting scheme differ across years. As a result, the index levels are not strictly comparable over time. To address this limitation, the subsequent analysis adopts a merged-corpus fitting approach to construct a time series index that is comparable across years.



Second, based on the merged-corpus TF-IDF fitting approach used to construct a comparable time-series index, we further introduce the Operational Distress Index to capture pressure and uncertainty at the operational level of the firm. The corresponding results are shown in the figure.

During the period from 2012 to 2016, the Hard Distress index exhibited a clear upward trend, reaching a local peak in 2015 and remaining at a relatively elevated level in 2016. This pattern is consistent with the company's sustained operational pressures and financial constraints during that period, indicating that risk disclosures related to solvency, financing risk, and capital structure stability were indeed intensified in the distress phase. At the same time, the Operational Distress Index also rose significantly and peaked in 2015, suggesting that the company not only strengthened its disclosure of financial pressures but also more frequently emphasized uncertainties related to product sales, customer demand, profitability, and operational stability. In contrast, although the Soft Distress index fluctuated during this period, it remained persistently high overall, reflecting the continued presence of macroeconomic and operational risks throughout the distress phase.

Entering the COVID-19 shock period in 2020, all indices again displayed notable changes. The Hard Distress index reached its highest level in 2021 across the full sample, indicating that during the pandemic shock the company more frequently employed language related to financing constraints, liquidity pressures, and debt arrangements. This suggests a marked increase in management's attention to potential financial risks. Subsequently, both the Soft Distress and Operational Distress indices rose steadily during 2022–2024, reflecting the sustained impact of post-pandemic macroeconomic volatility and industry cycle adjustments on corporate operational uncertainty.



5.2 Distress Indices and Future Stock Performance

To examine whether distress-related language predicts subsequent stock price declines, we test the relationship between the three distress indices (Hard, Soft, and Operational) and AMD's future 12-month abnormal returns.

5.2.1 Spearman Correlation Results

The Spearman correlations show that the three distress indices are related to future stock returns in different ways.

The Soft Distress index is negatively correlated with future abnormal returns **($\rho = -0.42$)**, consistent with the hypothesis that intensified distress language precedes stock price declines. In contrast, the Hard Distress index **($\rho = +0.31$)** and the Operational Distress index **($\rho = +0.16$)** both exhibit positive correlations with future returns, which is inconsistent with the hypothesis. Notably, the Soft index has the largest absolute correlation coefficient, suggesting that it contains relatively stronger predictive information compared to the other two indices.

5.2.2. High vs. Low Distress Group Comparison

We further divide the sample into high-distress and low-distress groups based on the median value of each index.

For the Hard Distress index, high-distress years are associated with an average future return of **0.263**, while low-distress years correspond to **0.152**. This suggests that years with stronger hard-risk language are followed by higher subsequent returns, indicating a reversal pattern rather than a stock price decline effect.

For the Soft Distress index, high-distress years are followed by an average future abnormal return of **−0.036**, whereas low-distress years exhibit **0.452**. The difference between the two groups is substantial. Economically, this result supports our hypothesis that greater intensity of risk language is associated with poorer subsequent stock performance.

For the Operational Distress index, high-distress years correspond to a future return of **0.309**, while low-distress years are associated with **−0.402**. Once again, the direction is opposite to our hypothesis, suggesting that elevated operational distress language may reflect a post-distress rebound or turnaround signal rather than an impending stock price decline.

5.2.3. OLS Regression Results

We next estimate simple OLS regressions of future abnormal returns on each distress index.

The Hard Distress coefficient is positive **(β = +15.3)** and statistically insignificant **(p = 0.526, R^2 = 0.034)**, indicating no predictive power and a direction inconsistent with the hypothesis.

The Soft Distress coefficient is negative **(β = −74.96)** and economically large, with a higher **R^2 (0.093)** than the Hard index. Although the result is not statistically significant **(p = 0.289)**, likely due to the small sample size, the direction of the effect is consistent with the hypothesis.

The Operational Distress coefficient is positive **(β = +50.96)** with the highest R^2 among the three **(0.128)**, yet remains statistically insignificant **(p = 0.210)**. The positive sign suggests that elevated operational distress language may be interpreted by the market as a potential turnaround or recovery signal rather than impending decline.

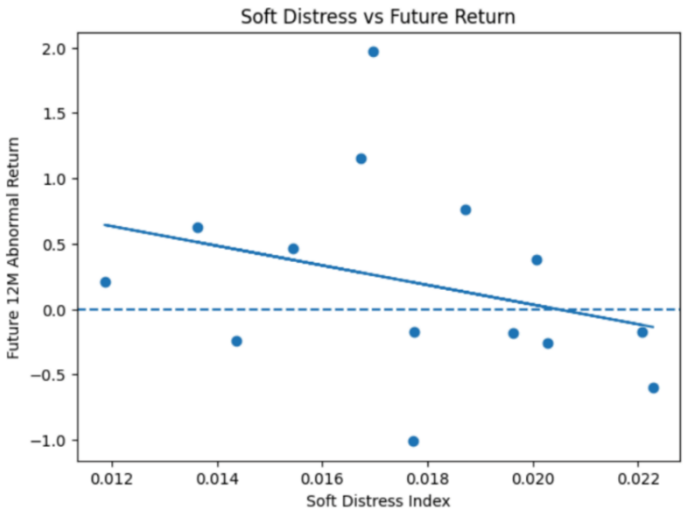
Although the Soft Distress index does not reach statistical significance in the regression analysis, it consistently exhibits the expected negative direction across correlation analysis, group comparisons, and regression results. We therefore focus on this measure in the subsequent visualization and interpretation, while exercising appropriate caution in statistical inference.

Index	Coefficient (β)	p-value	R^2	Direction vs Hypothesis
Hard Distress	15.3	0.526	0.034	Opposite
Soft Distress	-74.96	0.289	0.093	Consistent
Operational Distress	50.96	0.21	0.128	Opposite

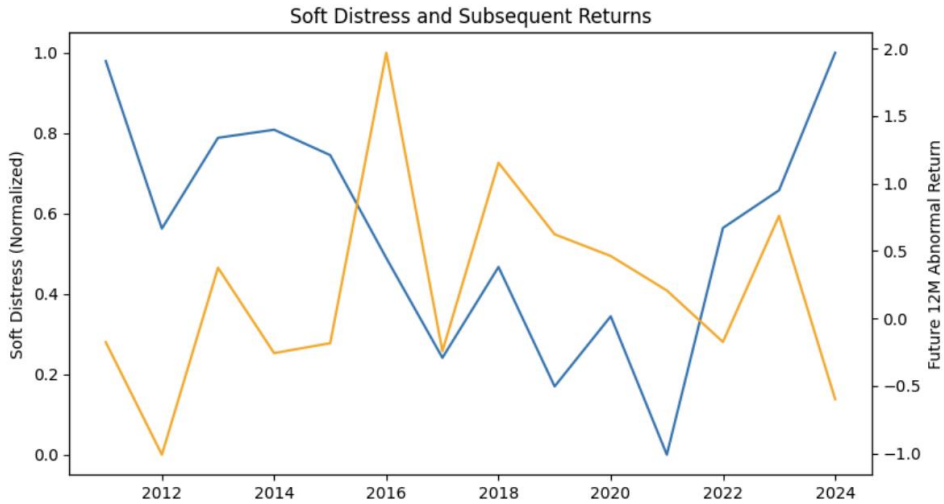
5.3 Soft Distress and Subsequent Returns

To further illustrate the predictive characteristics of the Soft Distress index, we provide additional visual analyses, including a scatter plot, a time-series comparison, and a high-versus-low grouping analysis.

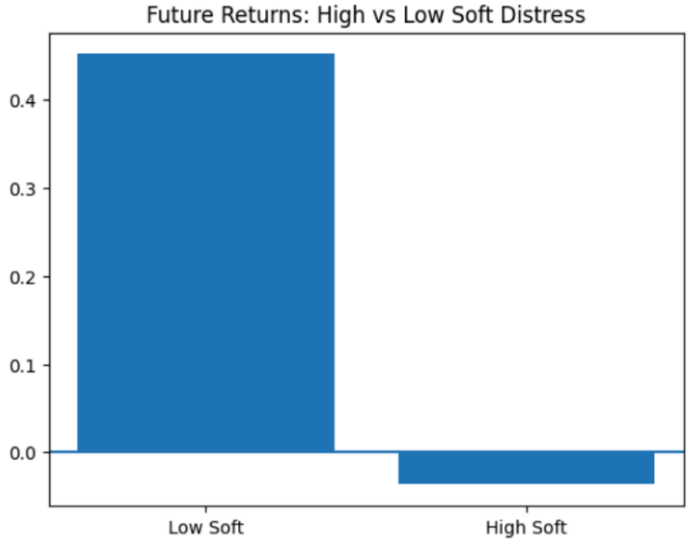
The scatter plot reveals a clear negative slope between the Soft Distress index and future 12-month abnormal returns. Although the sample size is small (N = 14), the downward trend line visually corroborates the earlier Spearman correlation results. Overall, higher levels of Soft Distress are generally associated with lower subsequent abnormal returns.



The time-series comparison shows that the Soft Distress index and future 12-month abnormal returns exhibit an overall inverse co-movement over the sample period. During 2012–2016, the Soft Distress index remained relatively elevated in most years, while subsequent returns were generally lower. Similarly, when the Soft Distress index rose again in 2023–2024, future returns weakened significantly. Although individual years display volatility and outliers, the overall time-series pattern provides directional evidence that higher levels of soft distress language may precede weaker subsequent stock performance.



The high-versus-low grouping figure shows that the Low Soft Distress group has a substantially positive average future abnormal return (approximately 0.45), whereas the High Soft Distress group exhibits a slightly negative average future abnormal return (approximately −0.04). The difference between the two groups is considerable and clearly indicates an inverse relationship: when the Soft Distress index is high, the company's stock performance in the following year tends to be weaker; when the Soft Distress index is low, subsequent returns are relatively stronger.



6. Discussion, Economic Interpretation, and Limitations

6.1 Economic Interpretation

Among the three distress dimensions, only Soft Distress exhibits the expected negative association with future abnormal returns, while Hard and Operational Distress display countercyclical patterns. This divergence can be understood in the context of AMD's structural characteristics as a turnaround firm.

During the 2012–2015 distress phase, increases in Hard and Operational distress language largely reflected restructuring troughs and strategic realignment rather than signals of further deterioration. In such settings, intensified disclosures of extreme financial risks and operational pressures tend to coincide with cyclical bottoms, followed by recovery and market revaluation, thereby generating positive subsequent returns.

In contrast, Soft Distress captures broader and more persistent signals such as cash flow pressure, earnings weakness, and macro-level uncertainty. This type of language is more likely to be interpreted by the market as forward-looking negative information, which explains its consistent inverse relationship with subsequent abnormal returns. Overall, different dimensions of distress language carry distinct informational meanings in asset pricing.

6.2 Limitations and Future Research

It is important to emphasize that although the Soft Distress index exhibits a negative relationship consistent with the hypothesis in both the correlation analysis and the high-versus-low grouping comparison, its coefficient in the OLS regression does not reach statistical significance ($p > 0.05$). Therefore, the empirical findings of this study provide support primarily in terms of directional and economic relevance rather than statistical confirmation.

The lack of statistical significance is mainly attributable to structural constraints of the sample rather

than methodological shortcomings. The sample consists of only 14 annual observations, resulting in limited statistical power. With such restricted degrees of freedom, achieving statistical significance in time-series regressions is inherently difficult. In addition, this study focuses on a single firm, and the observed patterns may reflect AMD-specific turnaround characteristics rather than a general market phenomenon.

Future research could improve statistical power and external validity by expanding both the sample dimension and the textual scope. For instance, at the textual level, incorporating Item 7 (Management's Discussion and Analysis) may help capture more detailed discussions of operational conditions and cash flow pressures. At the firm level, extending the analysis to peer firms would allow for industry-level comparisons and more effectively distinguish firm-specific distress language from broader macroeconomic risk disclosures.

6.3 Conclusion

This study examines whether distress-related risk language disclosed in AMD's Item 1A (Risk Factors) intensifies prior to periods of stock price decline. The empirical findings indicate that the relationship between different dimensions of distress language and future returns is not uniform. In particular, the Soft Distress index exhibits a consistent negative association with future 12-month abnormal returns and tends to remain elevated during periods preceding weaker stock performance. This suggests that broader and more persistent forms of operational and uncertainty-related risk language may serve as early signals of subsequent downside risk. However, it should be noted that although the Soft Distress coefficient in the OLS regression has the expected sign, it does not reach statistical significance; therefore, the findings provide support primarily in terms of direction and economic relevance rather than statistical confirmation.

In contrast, the Hard and Operational Distress indices do not display a significant negative predictive relationship. In the context of AMD as a turnaround firm, extreme financial distress and restructuring-related language often coincides with cyclical troughs that are followed by recovery, thereby generating countercyclical return patterns. Thus, whether distress risk language intensifies before stock price declines appears to depend on both the type of distress and the firm's position in the business cycle.

Overall, the evidence provides partial support for the research question. While distress risk language does not uniformly intensify across all dimensions, softer and more persistent forms of distress language appear to exhibit stronger forward-looking relevance in predicting subsequent stock underperformance.

Appendix A

A1. Hard Distress

- going concern substantial doubt
- event of default default under debt
- debt covenant compliance financial covenant
- breach violate violation not in compliance covenant
- covenant waiver waived waiver of covenant
- refinancing credit facility amended credit agreement maturity extended
- revolving credit facility revolver
- goodwill impairment write down intangible assets impairment
- restructuring charges workforce reduction cost reduction

A2. Soft Distress

- loss
- net loss
- negative operating income operating loss
- cash flow pressure
- liquidity constraints
- cash burn
- net cash used in operating activities
- decline in cash flows from operations operating cash flow decline

A3. Operational Distress

- operating loss
- net loss
- negative cash flow
- decline in revenue
- decrease in revenue
- revenue decline
- decline in gross margin
- margin erosion
- competitive pressure
- loss of market share
- pricing pressure
- restructuring
- cost reduction
- workforce reduction
- impairment charge

Appendix B

- breach of covenant
- covenant waiver

- debt covenant compliance
- default on debt
- going concern substantial doubt
- goodwill impairment
- liquidity risk
- negative cash flow
- operating cash flow decline
- refinancing credit facility
- restructuring charges
- revolving credit facility

Appendix C

Query	**Precision@3**
default on debt	0.75
goodwill impairment	0.75
revolving credit facility	0.625
liquidity risk	0.5
restructuring charges	0.125
debt covenant compliance	0
breach of covenant	0
covenant waiver	0
negative cash flow	0
going concern substantial doubt	0
refinancing credit facility	0
operating cash flow decline	0